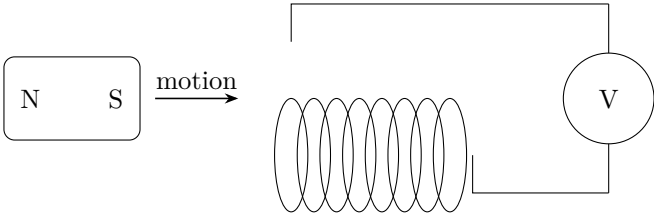


ONLY THE HAND-MOVED SOURCE DESCRIBED HERE

Use a hand-moved permanent magnet and a passive coil connected to a meter. Do not substitute a powered electromagnet, motor, transformer, outlet-powered apparatus, ignition component, large magnet assembly, or high-voltage device. Keep strong magnets away from implants and magnetic storage. **STOP: no outlets, mains leads, wall adapters, large battery packs, large capacitors, ignition coils, or exposed high voltage.**

Make the detector

Wind 200–300 turns of fine enamelled wire around a cardboard tube; tape the coil, scrape the two ends, and connect it to a digital multimeter on its most sensitive DC voltage range. The coil has no battery.



Run the sign test

Zero is a result. Hold the magnet stationary inside the coil; record the meter. Then make four controlled moves, using the same travel distance:

Move	Peak sign	Peak magnitude	Repeats agree?
North pole in, slow			
North pole out, slow			
South pole in, slow			
North pole in, fast			

Evidence record	
Prediction	
Changed variable	
Measured result	
Claim + because	

Build the compact model

A *change* in magnetic flux through the coil induces an emf. Reversing motion or magnet orientation reverses the sign. Faster change generally makes a larger magnitude. The induced effect opposes the change that produces it: this is the qualitative Faraday–Lenz model.

Where the energy comes from

The meter response is not free energy. Your hand does mechanical work against the induced interaction; some energy transfers electrically and ultimately becomes thermal energy. A generator scales this conversion with controlled motion, magnetic fields and circuits.

A fairer extension

Compare 100 and 300 turns using the same magnet, tube, travel distance and a metronome to time the stroke. Make five trials per coil and compare medians. Then name the largest uncontrolled variable. Do not compare hand speed by memory.

What this does—and does not—show

It supports a relationship among changing flux, polarity and induced emf. It does not establish the waveform of grid AC, transformer safety, or permission to connect a coil to household power.

Evidence: DOE S06. Safety boundary: OSHA S01–S03. See the provider-neutral electricity source register.

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